Hence, the negative ions could be chlorine ions (Cl^-).

Q6(a)
The binding energy per nucleon of the α-particle is significantly higher than the deuterium nucleus and is particularly stable for its mass number and hence more likely to be produced.

6(b)
Mass number $A = 224 - 4 = 220$

Proton number $Z = 88 - 2 = 86$

Neutron number $= 220 - 86 = 134$

6(c)
Due to the conservation of mass-energy, the difference in mass is converted into the kinetic energy of the products of the decay.

6(d)
Energy released = [Mass reactants – mass products]c^2
$= [224.0202 - (220.0114 + 4.0026)] \times 1.66 \times 10^{-27} \times (3.0 \times 10^8)^2$
$= 9.2628 \times 10^{-13}$ J
= 5.789 MeV

$K_\alpha + K_{Rn} = 5.789$ MeV

Hence, energy of α-particle = 5.789 – 0.1037 = **5.686 MeV**

6(e)

$$N = \frac{5.686\text{x}10^6}{31} = 1.83\text{x}10^5 \text{ ionisations}$$

Since the range of α-particles is about 3 cm (Accept 2 to 5 cm)

No. of ion-electron pairs per unit length $= \dfrac{1.83\times10^5}{3\times10^{-2}} = 6.1\times10^6 \text{ m}^{-1}$

since one ionisation produces an ion-electron pair.

Q7(a)